Mid-II (MVC)

Rubric + Marking Scheme

Total = 65

Solution # 1(a)

Total = 10 Marks

Given : $\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{(1+x^2+y^2)^2}\,dy\,dx.$

- **Step 1 : Identify the region**

$y = -\sqrt{1-x^2}$ to $\sqrt{1-x^2}$, $x = -1$ to 1 (Limits)

$y = \sqrt{1-x^2} \Rightarrow y^2 = 1-x^2$ → 3

$x^2+y^2 = 1$. (unit circle – centered at the origin, radius 1)

The region is symmetric which makes polar conversion straight-forward

- **Step 2 : Convert the integrand to polar coordinates**

$x = r\cos\theta$, $y = r\sin\theta$, $dx\,dy = r\,dr\,d\theta$ (polar coordinate formulas)

So,

$\Rightarrow \dfrac{2}{(1+x^2+y^2)^2} = \dfrac{2}{(1+r^2)^2}$ $(\because r^2 = x^2+y^2)$ → 2

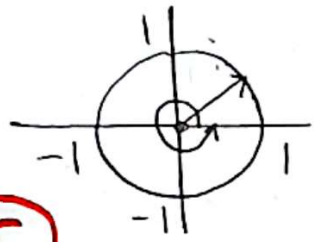
$\Rightarrow dy\,dx = r\,dr\,d\theta.$

- **Step 3 : Set up Polar Integral**

$r : 0 \to 1$

$\theta : 0 \to 2\pi$

$$\int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=1} \frac{2}{(1+r^2)^2}\, r\,dr\,d\theta.$$



→ 5

This is the standard polar form.

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Other Equivalent Ways:

① Full circle ($\theta: 0 \to 2\pi$).

$$\int_0^{2\pi}\int_0^1 \frac{2}{(1+r^2)^2}\, r\,dr\,d\theta$$

(Standard form)

② Half circle ($\theta: 0 \to \pi$).

$$2\int_0^{\pi}\int_0^1 \frac{2}{(1+r^2)^2}\, r\,dr\,d\theta.$$

(Option 1)

③ Quarter circle ($\theta: 0 \to \pi/2$)

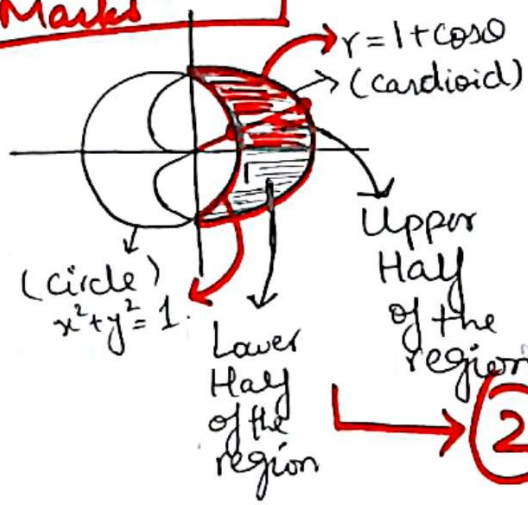
$$4\int_0^{\pi/2}\int_0^1 \frac{2}{(1+r^2)^2}\, r\,dr\,d\theta$$

(Option 2).

Total Marks = 10

* Solution 1(b)

Given : Region that lies inside $r = 1+\cos\theta$ and outside the circle $r=1$



→ 2

- Step 1: Set up the double integral

$$A = \iint_{\text{region}} r\,dr\,d\theta$$

- Inner curve : $r = 1$
- Outer curve : $r = 1+\cos\theta$
- Upper half of the region : $\theta = 0 \to \frac{\pi}{2}$ (due to symmetry about the x-axis)

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$$A = 2\int_0^{\pi/2}\int_0^{1+\cos\theta} r\,dr\,d\theta. \longrightarrow 2$$

- Step 2 : Integrate w.r.t 'r'

$$A = 2\int_0^{\pi/2} \frac{r^2}{2}\Big|_0^{1+\cos\theta} d\theta = 2\int_0^{\pi/2}\left[\frac{(1+\cos\theta)^2}{2} - \frac{(1)^2}{2}\right] d\theta.$$

$$A = \cancel{2}\int_0^{\pi/2}\left[\frac{(1+\cos\theta)^2 - 1}{\cancel{2}}\right] d\theta = \int_0^{\pi/2}\left[(1+\cos\theta)^2 - 1\right] d\theta. \longrightarrow 2$$

- Step 3: Expand the integrand

$$A = \int_0^{\pi/2}\left[\cancel{1} + 2\cos\theta + \cos^2\theta - \cancel{1}\right] d\theta = \int_0^{\pi/2}\left[2\cos\theta + \cos^2\theta\right] d\theta$$

$$A = \int_0^{\pi/2}\left[2\cos\theta + \frac{1}{2} + \frac{\cos 2\theta}{2}\right] d\theta \quad \left(\because \cos^2\theta = \frac{1+\cos 2\theta}{2}\right)$$

- Step 4: Integrate w.r.t θ

$$A = \left[2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\pi/2} \longrightarrow 2$$

$$= \left(2 + \frac{\pi}{4}\right) - 0$$

- Step 5: Final Answer

$$\boxed{A = \frac{8+\pi}{4}} \longrightarrow 2$$

$\longleftrightarrow$

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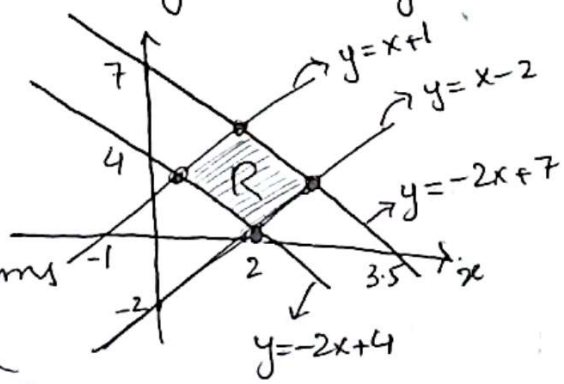
* Solution 1(c)

Total Marks = 10

Given: $u = x - y$, $v = 2x + y$ (Transformation)

R: $y = -2x + 4$, $y = -2x + 7$, $y = x - 2$, $y = x + 1$.

Step 1: Sketch the region R



Step 2: Find x & y values in terms of u & v

Here $y = -2x + 4$, $y = -2x + 7$

$y + 2x = 4$, $y + 2x = 7$

And $y = x - 2$, $y = x + 1$

$x - y = 2$, $x - y = -1$

We know $u = x - y$, $v = 2x + y$

$u + v = x - \cancel{y} + 2x + \cancel{y}$, $u = x - y$

$u + v = 3x$, $y = x - u$

$\boxed{x = \frac{u+v}{3}}$, $y = \frac{u+v}{3} - u = \frac{u+v-3u}{3}$

$\boxed{y = \frac{v-2u}{3}}$ → 2

Step 3: Find Jacobian: $J = \frac{\partial(x,y)}{\partial(u,v)}$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1/3 & 1/3 \\ -2/3 & 1/3 \end{vmatrix} = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}$$

$\boxed{|J| = \frac{1}{3}}$ → 2

$\boxed{\text{or}}$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{\frac{\partial(u,v)}{\partial(x,y)}}$$

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$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1+2=3$$ Pg# 5

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{\frac{\partial(u,v)}{\partial(x,y)}} = \frac{1}{3}$$

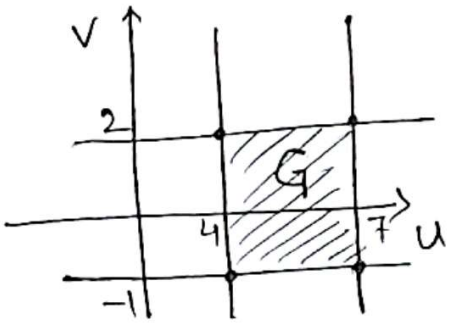
$$|J| = \frac{1}{3}$$

Step 4: New Region + Sketch

As $y+2x=4 \Rightarrow u=4$.

$y+2x=7 \Rightarrow u=7$

$\therefore \boxed{4 \le u \le 7}$



Also $x-y=2 \Rightarrow v=2$

$x-y=-1 \Rightarrow v=-1$

$\therefore \boxed{-1 \le v \le 2}$ $\longrightarrow$ 2

Step 5: Simplify the integrand

$$\iint_R (2x^2 - xy - y^2)\,dx\,dy$$

Take $2x^2 - xy - y^2 = 2\left[\frac{(u+v)^2}{9}\right] - \left[\frac{(u+v)(v-2u)}{9}\right] - \left[\frac{(v-2u)^2}{9}\right]$

$$= 2\left[\frac{u^2+2uv+v^2}{9}\right] - \left[\frac{uv-2u^2+v^2-2uv}{9}\right] - \left[\frac{v^2-4uv+4u^2}{9}\right]$$

$$= \frac{2u^2+4uv+2v^2}{9} - \left[\frac{-2u^2-uv+v^2}{9}\right] - \left[\frac{v^2-4uv+4u^2}{9}\right]$$

$$= \frac{(2u^2+4uv+2v^2)+(2u^2+uv-v^2)-(v^2-4uv+4u^2)}{9}$$

$$= \frac{9uv}{9} = uv$$ $\longrightarrow$ 1

$$\iint_R (2x^2 - xy - y^2)\,dx\,dy = \iint_G uv \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

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Step 6: Solve New Integral

$$\iint_R (2x^2 - xy - y^2)\,dx\,dy = \int_{-1}^{2}\int_{4}^{7} uv\left(\frac{1}{3}\right) du\,dv$$

$$= \frac{1}{3}\int_{-1}^{2}\left[\frac{u^2}{2}v\right]_4^7 dv$$

$$= \frac{1}{3}\int_{-1}^{2} v\left[\frac{7^2}{2} - \frac{4^2}{2}\right] dv$$

$$= \frac{1}{3}\int_{-1}^{2} v\left[\frac{33}{2}\right] dv$$

$$= \frac{11}{2}\int_{-1}^{2} v\,dv$$

$$= \frac{11}{2}\left.\frac{v^2}{2}\right|_{-1}^{2} = \frac{11}{4}\left[2^2 - (-1)^2\right]$$

$$= \frac{11}{4}[4-1] = \boxed{\frac{33}{4}}$$

$$\boxed{\therefore \iint_R (2x^2 - xy - y^2)\,dx\,dy = \int_{-1}^{2}\int_{4}^{7} uv\cdot\frac{1}{3}\,du\,dv = \frac{33}{4}} \longrightarrow 3$$

Total Marks = 10

★ Solution 1(d)

Given: $\rho = 2$ (sphere of radius 4)

or $x^2 + y^2 + z^2 = 4$

(a) Spherical Coordinates: (5 marks)

Volume element: $dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta \longrightarrow 2$

$0 \le \phi \le \pi$
$0 \le \theta \le 2\pi$.

Radius: $0 \le \rho \le 2$

Standard Form (Full sphere): $\boxed{V = \int_0^{2\pi}\int_0^{\pi}\int_0^{2} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta}$

Other Equivalent Forms:

① $\theta: 0 \to \pi$, $\phi: 0 \to \pi$.

$$\boxed{V = 2\int_0^{\pi}\int_0^{\pi}\int_0^{2} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta} \longrightarrow 3$$

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(2) $\theta : 0 \to \pi/2$, $\phi : 0 \to \pi$

$$V = 4\int_0^{\pi/2}\int_0^{\pi}\int_0^{2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta.$$

(3) $\theta : 0 \to 2\pi$, $\phi : 0 \to \pi/2$

$$V = \int_0^{2\pi} 2\int_0^{\pi/2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\int_0^{2\pi}\int_0^{\pi/2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta.$$

(4) $\theta : 0 \to \pi$, $\phi : 0 \to \pi/2$

$$V = 2\int_0^{\pi} 2\int_0^{\pi/2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 4\int_0^{\pi}\int_0^{\pi/2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

(5) $\theta : 0 \to \pi/2$, $\phi : 0 \to \pi/2$

$$\boxed{V = 4\int_0^{\pi/2} 2\int_0^{\pi/2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 8\int_0^{\pi/2}\int_0^{\pi/2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta}$$

(b) <u>Cylindrical Coordinates</u>: (5 marks)

Volume element : $dV = dz \, r \, dr \, d\theta$.

Radius : $r \geq 0$

θ : $0 \leq \theta \leq 2\pi$. , z : Height along z-axis

<u>Step 1: Convert sphere equation</u>

Sphere : $x^2 + y^2 + z^2 = 4$. (given).

In cylindrical : $x^2 + y^2 = r^2$

So $r^2 + z^2 = 4$. $\longrightarrow$ (1)

<u>Step 2: Determine Limits</u>

From : $r^2 + z^2 = 4$

Solve for z : $z^2 = 4 - r^2$

$z = \pm\sqrt{4 - r^2}$

$-\sqrt{4-r^2} \leq z \leq \sqrt{4-r^2}$ $\longrightarrow$ (1)

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Standard Form : $\boxed{V=\int_0^{2\pi}\int_0^{2}\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta}$ $\hookrightarrow$ 3

Other Equivalent Forms:

1. $\theta: 0 \to \pi$, $z: -\sqrt{4-r^2} \to \sqrt{4-r^2}$

$$V = 2\int_0^{\pi}\int_0^{2}\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta.$$

2. $\theta: 0 \to \pi/2$, $z: -\sqrt{4-r^2} \to \sqrt{4-r^2}$

$$V = 4\int_0^{\pi/2}\int_0^{2}\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta.$$

3. $\theta: 0 \to 2\pi$, $z: 0 \to \sqrt{4-r^2}$

$$V = \int_0^{2\pi}\int_0^{2} 2\int_0^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta = 2\int_0^{2\pi}\int_0^{2}\int_0^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta$$

4. $\theta: 0 \to \pi$, $z: 0 \to \sqrt{4-r^2}$

$$V = 2\int_0^{\pi}\int_0^{2} 2\int_0^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta = 4\int_0^{\pi}\int_0^{2}\int_0^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta.$$

5. $\theta: 0 \to \pi/2$, $z: 0 \to \sqrt{4-r^2}$

$$\boxed{V = 4\int_0^{\pi/2}\int_0^{2} 2\int_0^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta = 8\int_0^{\pi/2}\int_0^{2}\int_0^{\sqrt{4-r^2}} dz\,r\,dr\,d\theta.}$$

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* Solution 1(e)

Total Marks = 10

Given : $f(x,y) = ye^{x^2}$

$\vec{r}(t) = 4t\,\hat{i} - 3t\,\hat{j}$, $1 \le t \le 2$

Step 1: Line Integral Formula

$$\int_C f(x,y)\,ds = \int_a^b f(\vec{r}(t))\,|\vec{r}'(t)|\,dt = \int_a^b f(x(t),y(t))\cdot|\vec{r}'(t)|\,dt \longrightarrow 1$$

where $\vec{r}'(t) = \dfrac{d\vec{r}}{dt}$, $|\vec{r}'(t)|$ = speed of the curve / magnitude of $\vec{r}(t)$

Step 2 : Find $x(t)$ and $y(t)$

From $\vec{r}(t) = 4t\,\hat{i} - 3t\,\hat{j}$.

$x(t) = 4t$, $y(t) = -3t$.

Step 3 : Find $\vec{r}'(t)$ and $|\vec{r}'(t)|$

$$\vec{r}'(t) = \frac{d}{dt}(4t\,\hat{i} - 3t\,\hat{j}) = 4\hat{i} - 3\hat{j} \longrightarrow 4$$

$$|\vec{r}'(t)| = \sqrt{(4)^2 + (-3)^2} = \sqrt{16+9} = \sqrt{25} = 5$$

Step 4: Substitute $\vec{r}(t) = (x(t), y(t))$ into $f(x,y)$

From $f(x,y) = ye^{x^2}$

$$f(\vec{r}(t)) = f(x(t), y(t)) = (-3t)\,e^{(4t)^2} = -3te^{16t^2} \longrightarrow 1$$

Step 5: Solve the integral

$$\int_C f(x,y)\,ds = \int_1^2 -3te^{16t^2}(5)\,dt = -15\int_1^2 te^{16t^2}\,dt$$

Let $u = 16t^2$, $du = 32t\,dt$

$\dfrac{du}{32} = t\,dt$. $\longrightarrow 2$

When $t = 1 \rightarrow u = 16$.

When $t = 2 \rightarrow u = 64$.

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$$-15\int_1^2 t e^{16t^2}\,dt = -15\int_{16}^{64} e^u \frac{du}{32}$$

$$= -\frac{15}{32}\int_{16}^{64} e^u\,du$$

$$= -\frac{15}{32}\left[e^u\right]_{16}^{64} = -\frac{15}{32}\left[e^{64} - e^{16}\right]$$

$$\int_1^2 f(\vec{r}(t))\cdot|\vec{r}\,'(t)|\,dt = \boxed{\frac{15}{32}\left(e^{16} - e^{64}\right)} \longrightarrow 2$$

Total Marks = 15

* Solution 1(f) → Fill in the blanks

1. $dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$
2. $\phi = \frac{\pi}{4}$
3. $J(u,v,w) = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$
4. Constant (Not functions of other variables)
5. $x = r\cos\theta,\ y = r\sin\theta,\ z = z$
   $r^2 = x^2 + y^2,\quad \tan\theta = \frac{y}{x}$
6. $dA = r\,dr\,d\theta$
7. Non-zero
8. Average value of F over D $= \frac{1}{\text{volume of } D}\iiint_D F\,dV$
9. $J(r,\theta) = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = r$
10. r^2
11. Volume of the solid D
12. $dV = dz\, r\,dr\,d\theta$
13. $J(\rho,\phi,\theta) = \begin{vmatrix} x_\rho & x_\phi & x_\theta \\ y_\rho & y_\phi & y_\theta \\ z_\rho & z_\phi & z_\theta \end{vmatrix} = \rho^2 \sin\phi$
14. distance from the origin to the point
15. angle from the positive z-axis.
16. Polar coordinates (xy plane) or (Cylindrical coordinates)
17. $x = \rho\sin\phi\cos\theta$
    $y = \rho\sin\phi\sin\theta$
    $z = \rho\cos\phi$
18. $r = \rho\sin\phi$
    $z = \rho\cos\phi$
    $\theta = \theta$
19. Region → Stretched
    Area → Increased
20. Region → Compressed
    Area → Decreased

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